Wavefront deflections vs. thermal gradients.

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Consider two rays that enter the telescope dome on opposite sides of the pupil, across a diameter. Assume a planar wavefront arriving at the slit of the enclosure.
Each ray traverses a path length PL as it moves through the dome air. Assume the dome air has a thermal structure so that the average temperature along the two paths differs by delta-T.

This means the two rays see a different average (line-integrated) index of refraction delta-n. The difference in optical path length for the two rays is PL*delta-n. Across a primary mirror of diameter D this induces a wavefront tilt of theta=PL*delta-n/D. We could have seen this from dimensional analysis alone.

Let's put in some numbers. For the Rubin telescope we have D=8m, PL = 10m. The conversion from delta-T to delta-n is dn/dT~-1E-6 per K. Setting the deflection equal to 1 arcsec we can solve for delta-T:

The line-integrated difference in index obeys delta-n=5E-6*(8/10) = 4e-6. So to get less than one arcsec of wavefront deflection we need to have the line integral of temperatures differ by no more than 4K.

To meet the Rubin dome-seeing spec of 0.08 arcsec we need line-integrated temperature difference to be less than 0.3K. So a measurement objective of 0.1K is very sensible.

For a 10m pathlength of acoustic interrogation we'd see a travel time difference driven by change in average sound speed. Take that change to be half the gradient, so 0.15K.
That induces change in sound speed of 0.6m/s/K * 0.15 ~ 0.1 m/s. So difference in sound travel time is 10m/330 m/s vs. 10m/330.1 m/s or around 20 microseconds.

We can easily resolve this acoustically, it's 10X our measurement error.

The light traverses the primary-to-secondary z distance not once but 3 times, converging on the third pass. So thermal path is really more like 20m so requirement is perhaps 2x more stringent?

We should set up two parallel acoustic monitors, parallel to the optical axis on opposite sides of the mirror. This will measure differences in the line-integrated index.